

Reg No.: _____

Name: _____

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

Sixth Semester B.Tech Degree Supplementary Examination May 2023 (2019 Scheme)

Course Code: EET302**Course Name: LINEAR CONTROL SYSTEMS**

Max. Marks: 100

Duration: 3 Hours

PART A*Answer all questions, each carries 3 marks.*

Marks

- | | | |
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| 1 | Compare open loop and closed loop systems. Give an example for each | (3) |
| 2 | Obtain the transfer function of a lead compensator with the help of an electrical network. | (3) |
| 3 | Draw the unit step response of an under damped system and mark various time domain specifications | (3) |
| 4 | State and explain Routh- Hurwitz stability criterion. | (3) |
| 5 | Write the design steps of lag compensator using root locus. | (3) |
| 6 | Explain the effect of adding poles and zeros to the nature of root locus. | (3) |
| 7 | What is the relationship between phase cross over frequency and gain cross over frequency for a stable system. | (3) |
| 8 | Point out the difference between minimum phase systems and non-minimum phase systems | (3) |
| 9 | State and explain Nyquist Stability Criterion. | (3) |
| 10 | Write procedure for designing a lead compensator design using Bode plot | (3) |

PART B*Answer one full question from each module, each carries 14 marks.***Module I**

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|----|--|-----|
| 11 | a) Derive the transfer function of an armature controlled DC motor and represent the system in block diagram form. | (7) |
| | b) Derive the transfer function of a lag-lead compensator. | (7) |

OR

- | | | |
|----|--|-----|
| 12 | a) Explain the following control system components | (8) |
| | (i) Sychro-transmitter | |
| | (ii) AC tachogenerator | |

- b) Obtain the transfer function of lag compensators using R-C circuit components and hence explain the characteristics of the lag compensators (6)

Module II

- 13 a) Derive the expression for unit step response of an under damped second order system. (8)
- b) A unity feedback control system is characterized by an open loop transfer function given below. Determine the value of K so that its damping ratio of the closed loop system is 0.5. (6)

$$G(s) = \frac{K}{s(s + 10)}$$

OR

- 14 a) Derive the expression for rise time and peak time of an under damped second order system. (8)
- b) Using Routh – Hurwitz Criterion ascertain the stability of the system whose characteristic equation is. (6)

$$s^6 + 3s^5 + 5s^4 + 9s^3 + 8s^2 + 6s + 4 = 0$$

Module III

- 15 a) Sketch the root locus of the system having open loop transfer function (14)
- $$G(s)H(s) = \frac{K}{s(s+2)(s+4)}$$
- Find the value of K so that the damping ratio of the closed loop system is 0.5.

OR

- 16 a) Design a phase lag compensator for a system whose open loop transfer function (14)
- $$G(s)H(s) = \frac{K}{s(s+4)(s+5)}$$
- to meet the following specifications

(i) Damping ratio = 0.707 (ii) $K_v \geq 5$ (iii) $\omega_n = 2$ rad/sec.

Module IV

- 17 a) Determine the gain cross over frequency, phase cross over frequency, gain margin and phase margin for the system with open loop transfer function (10)

$$G(s) = \frac{1}{s(2s + 1)(1 + s)}$$

- b) What is resonant frequency and resonant peak for a second order system? What is the relation between resonant peak of a system with the damping factor, η ? (4)

OR

- 18 Draw Bode Plot for a system whose open loop transfer function is (14)

$$G(s)H(s) = \frac{10}{s(1+s)(1+0.02s)}$$

Also Determine

- i. Gain Margin
- ii. Phase Margin
- iii. Closed Loop Stability

Module V

- 19 Draw the Nyquist plot for the system whose open loop transfer function is (14)

$$G(s) = \frac{K}{s(s+2)(s+10)}$$

Determine the range of K for which the system is stable

OR

- 20 A unity feedback system has an open loop transfer function (14)

$$G(s) = \frac{K}{s(1+2s)}$$

Design a suitable lag compensator so that phase margin is 45 degree and velocity error constant is 5.
